ABISEKHA MELFIN V

AI/ML ENGINEER | GENERATIVE AI | COMPUTER VISION | PYTHON & DJANGO | FULL STACK DEVELOPMENT

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Nationality: Indian | Visa Status: Open to relocation — UAE work visa / employer sponsorship required | Available immediately

PROFESSIONAL SUMMARY

AI/ML Engineer with 3+ years building applied AI systems — computer vision (OpenCV, YOLO, MediaPipe), Generative AI (RAG, pgvector semantic search, LLM-powered applications) and Python/Django backends to serve them. Built and shipped 6+ AI/computer-vision solutions in production, including HireAI, a recruitment platform using vector embeddings for semantic candidate-job matching and RAG-grounded interview generation. Comfortable owning the full pipeline: model/technique selection, backend API design (Django REST Framework), and integration into a working React frontend. Immediately available and open to relocating to Dubai / UAE on a full-time, employer-sponsored basis.

KEY SKILLS

- AI & Machine Learning: Machine Learning, Deep Learning, Computer Vision, OpenCV, YOLO, TensorFlow, Keras, Scikit-learn, MediaPipe
- Generative AI: LLM-powered applications, RAG (Retrieval-Augmented Generation), pgvector semantic search, prompt engineering, NLP-based resume/job matching
- AI Agents & Automation: n8n workflow automation, AI agent design, structured outputs, human-in-the-loop approval workflows, Gmail/API tool integration, prompt engineering for agentic workflows
- Programming Languages: Python, Java, JavaScript, SQL, HTML, CSS
- Web Frameworks & Libraries: React.js, Django, Django REST Framework, Spring Boot, jQuery, Bootstrap
- AR/VR: Unity, C#, AR Foundation, ARCore, Meta Quest 3 development
- Databases & Tools: SQL, MySQL, PostgreSQL, Git, GitHub, Jupyter Notebook, Google Colab, Figma, Adobe Photoshop
- Deployment & DevOps: Vercel, Render, Netlify, CI/CD with GitHub Actions, environment configuration

PROFESSIONAL EXPERIENCE

IT Engineer — IPCS Global Solutions, Tirunelveli *Apr 2025 – Present*

- Built and implemented 6+ AI/computer-vision solutions, including OCR-based document verification, driver drowsiness detection, helmet detection and AI-assisted interview evaluation.
- Designed and deployed an n8n-based AI agent (EmployeeAssist AI) that drafts official employee emails, routes them through a manager approval step with structured decisioning, and automatically notifies employees of the outcome via Gmail integration.
- Delivered 10+ software, AI and data-analytics projects using Python, Django, React and JavaScript, contributing across planning, development, testing, deployment and documentation.
- Created and maintained technical documentation and project reports supporting 6+ final-year projects and multiple academic/industry engagements.
- Collaborated with technical teams, students, academic stakeholders and clients across 10+ concurrent software, AI and training initiatives.
- Supported 100+ internship participants and 6 final-year projects, and contributed to interview preparation and placement activities resulting in approximately 48 student placements.

AR/VR Developer & AI Technology Specialist — Netcom Computers, Tirunelveli *Feb 2024 – Apr 2025*

- Developed 5+ interactive AR/VR and AI technology projects using Unity, C#, Python and computer-vision technologies.
- Built an AR indoor navigation application covering a 3-floor environment using Unity and AR Foundation/ARCore.
- Developed an immersive VR “abandoned sea castle” experience for Meta Quest 3, focusing on interactive 3D environments and user experience.
- Integrated AI and computer-vision techniques using Python, OpenCV, YOLO and MediaPipe into interactive application prototypes.
- Managed the full build-test-deploy cycle across AR/VR and AI application development, including implementation, debugging and deployment.

Graphic & Web Design Intern — Capdigisoft Solutions, Coimbatore *Sep 2023 – Mar 2024*

- Built 8+ responsive websites and web interfaces using HTML, CSS and JavaScript.
- Designed UI/UX layouts, visual assets and website interfaces for 12+ web projects, focusing on usability, responsive design and visual consistency.
- Contributed to web design, frontend development and client-oriented digital projects.

KEY PROJECTS (GITHUB.COM/ABISEKHA02)

EmployeeAssist AI — n8n AI Agent for Employee Request Automation | *n8n, OpenAI, Gmail API, AI Agents, Structured Output Parsing*

- Built an AI agent workflow in n8n that converts a single chat message into a formal employee request (leave, permission, approval or announcement), using an OpenAI-backed agent with conversation memory.
- Implemented structured output parsing so the agent's response drives workflow logic (status, recipient, email content) instead of free-form text.
- Added a human-in-the-loop manager approval step using Gmail's Send-and-Wait operation with Approve/Decline actions, followed by conditional routing (IF logic) to notify the employee of the outcome.

HireAI — AI-Powered Resume Screening & Hiring Assistant | *React, TypeScript, Django REST Framework, PostgreSQL, pgvector, LLM/RAG*

- Built an end-to-end AI recruitment platform for resume parsing, semantic candidate-job matching, interview generation and AI-based answer evaluation.
- Implemented vector embeddings with pgvector rather than keyword-only matching, enabling semantic comparison between candidate profiles and job requirements.
- Developed RAG-grounded interview generation using parsed candidate information and job requirements.
- Implemented 3 user roles — recruiter, candidate and admin — with JWT authentication and role-based access control.
- Deployed the production application using Vercel and Render.

AI-Based Helmet Detection System | *Python, OpenCV, YOLO*

- Built a real-time computer-vision system for helmet detection using YOLO and OpenCV.
- Implemented the detection pipeline for practical traffic-safety monitoring scenarios.

Repository: github.com/abisekha02/AI-Helmet-Detection

Gesture-Controlled Virtual Mouse | *Python, OpenCV, MediaPipe*

- Built a touchless cursor-control application using MediaPipe hand-landmark detection and gesture recognition.
- Implemented multiple hand gestures mapped to mouse-control actions using Python and OpenCV.

Repository: github.com/abisekha02/Gesture-Controlled-Virtual-Mouse

OCR-Based Document Verification System | *Python, OCR, Computer Vision*

- Built an AI-powered OCR system for extracting and verifying information from ID documents and academic certificates.
- Automated document-data extraction and validation to reduce manual verification effort.

Repository: github.com/abisekha02/OCR-Based-Document-Verification-System

CareerFlow — Job Application & Interview Tracker | *React, TypeScript, Tailwind CSS, Django REST Framework, PostgreSQL, JWT*

- Built a full-stack job-search platform designed to manage 300+ job applications and recruitment activities.
- Developed a JWT-authenticated REST API covering application, interview and document management.
- Built dashboard analytics for tracking application activity, outcomes and job-search trends.
- Implemented responsive React components with TypeScript and integrated PostgreSQL-backed Django APIs.
- Deployed the frontend and backend using Vercel and Render.

Live Demo: careerflow-opal.vercel.app

EDUCATION

B.E., Computer Science and Engineering — Vaigai College of Engineering, Madurai — 89.95%
Higher Secondary Certificate — St. Ignatius Convent Higher Secondary School, Tirunelveli — 90%
SSLC — St. Ignatius Convent Higher Secondary School, Tirunelveli — 89%

CERTIFICATIONS

- Big Data and Machine Learning Fundamentals — Google Cloud
- Foundations of User Experience (UX) Design — Google
- Create High-Fidelity Designs and Prototypes in Figma — Google
- Graphic Design Foundations — LinkedIn Learning